10. A sector AOB, of a circle centre O, has radius r cm and angle θ radians.

Given that the area of the sector is $6\,\text{cm}^2$ and that the perimeter of the sector is $10\,\text{cm}$,

(a) show that

$$3\theta^2 - 13\theta + 12 = 0$$

(4)

(b) Hence find possible values of r and θ.

(3)